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NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Seventh Semester B.E. (CCE) (Credit System) Degree Examinations

Supplementary Examinations - December 2025

21CCE111 - INTRODUCTION TO BLOCKCHAIN TECHNOLOGY

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I** & **Unit – II** each and **One full** question from **Unit – III**.*

Unit – I		Marks	BT*	CO*	PO*
1.	a) Identify and describe the key features of blockchain technology, explaining how each feature contributes to its overall functionality.	10	L*2	1	2,3
	b) Briefly explain the following blockchain concepts: i. Public Blockchains ii. Private Blockchains iii. Semi-private Blockchains iv. Sidechains v. Tokenized Blockchains.	10	L2	1	2,3
2.	a) Identify the key components of a blockchain system that require decentralization to achieve a fully decentralized platform. Explain with the help of diagram for clarity.	10	L2	1	2,3
	b) Explain in detail the two methods of decentralization used in blockchain technology.	5	L2	1	2,3
	c) Explain the operations of a stream cipher with the help of a diagram.	5	L2	2	2,3
3.	a) Using diagrams explain electronic code block (ECB) and cipher block chaining (CBC) block cipher modes of operation function and describe their effects on data security and performance.	10	L2	2	2,3
	b) Outline the key differences between DES and AES in terms of their encryption algorithms?	5	L2	2	2,3
	c) Explain the importance of two practical and three security properties in hash functions for cryptography and security applications.	5	L2	2	2,3
Unit – II					
4.	a) Imagine you are part of an online marketplace where users can buy and sell digital assets. You need to create a contract that governs the transaction of these assets. Which contract type would you choose, a smart contract or a Ricardian contract, if: i. You want the contract to be automatically executed once specific conditions are met (e.g., when payment is made, the asset is transferred instantly)? ii. You need the contract to be legally enforceable in traditional courts of law, while also ensuring that both parties can understand the terms of the contract in human-readable language? iii. You want the contract to be fully decentralized and trustless, relying only on code to execute the agreed-upon terms without intermediaries? iv. You want the contract to incorporate legal language so that it can serve as both a technical specification and a legally binding agreement?	4	L3	3	2

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| b) | Summarize the properties of Ricardian contracts that enable them to act as both legally binding agreements and automatically executable digital contracts. | 6 | L2 | 3 | 2 |
| c) | Explain a simplified model of an oracle interacting with a smart contract on the blockchain. | 10 | L2 | 3 | 2 |
| 5. a) | Discuss the Ethereum Virtual Machine and its operation, providing a diagram to illustrate its functions. | 10 | L2 | 3 | 2 |
| b) | Develop a smart contract using the Voting Manager template that manages the voting process with different states. Your implementation should include: | | | | |
| i. | State Initialization: Ensure the contract initializes with the voting status set to notStarted. | | | | |
| ii. | State Transition: Implement a function to start the voting process, which changes the status to ongoing and increments a counter. | | | | |
| iii. | State Termination: Create a function to end the voting process, which sets the status to ended. | | | | |
| iv. | Status Retrieval: Implement a function that returns a human-readable string representing the current voting status. | 10 | L3 | 4 | 2,3 |
| 6. a) | Explain the concept of functions in Solidity, including function types (view, pure), modifiers, and return types. | 6 | L3 | 4 | 2,3 |
| b) | Implement a smart contract called Simple Mapping that uses a mapping to store and retrieve ages associated with names. | 8 | L3 | 4 | 2,3 |
| c) | Explain the use of mappings and arrays in Solidity, including their declaration, initialization and usage. | 6 | L2 | 4 | 2,3 |

Unit – III

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| 7. a) | Explain the reference architecture of Hyperledger Fabric and describe its key components and how they interact to enable modular and scalable blockchain solutions. | 10 | L2 | 5 | 2 |
| b) | Describe the key tools and frameworks provided by Hyperledger and their respective roles in supporting enterprise blockchain solutions. | 10 | L2 | 5 | 2 |
| 8. a) | Explain the essential building blocks of blockchain solutions. | 10 | L2 | 5 | 2 |
| b) | Show the advantages that open-source technologies, such as Hyperledger and its suite of projects, offer to the blockchain industry. | 10 | L2 | 5 | 2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
